**Date:** May 25th, 2026

**Week 0:** Orientation, Onboarding & Python Readiness

# CariSurg Healthcare AI Program

## Assignment 6: At-Risk Patient Flagging Rule

### Pseudocode for At-Risk Rule

```
FOR each patient in dataset:

  SET at_risk = FALSE

  IF SBP < 90:
    at_risk = TRUE   # hypotension

  IF DBP < 60:
    at_risk = TRUE   # poor perfusion indicator

  IF pulse < 50 OR pulse > 120:
    at_risk = TRUE   # bradycardia or tachycardia

  IF RR < 8 OR RR > 24:
    at_risk = TRUE   # respiratory distress or suppression

  IF Temp < 35 OR Temp > 38.3:
    at_risk = TRUE   # hypothermia or fever/infection

  IF SpO2 < 92:
    at_risk = TRUE   # hypoxemia

  IF GCS < 13:
    at_risk = TRUE   # altered consciousness

  RETURN at_risk
```

# 3. Clinical Justification of Thresholds

This rule uses **widely accepted early warning thresholds** used in emergency medicine and hospital triage systems (e.g., NEWS2-style logic).

- **SBP < 90 mmHg** → indicates hypotension and possible shock or poor perfusion.
- **DBP < 60 mmHg** → suggests reduced vascular resistance or circulatory compromise.
- **Pulse:** Bradycardia (<50 bpm) or tachycardia (>120 bpm) may indicate cardiac dysfunction, infection, or physiological stress.
- **Respiratory Rate:** Values below 10 or above 24 breaths per minute may signal respiratory failure, sepsis, or acute distress.
- **Temperature:** Hypothermia (<36°C) or fever (>38.5°C) can indicate infection, inflammation, or systemic illness.
- **Oxygen Saturation ($SpO_2$):** Values below 94% suggest hypoxaemia and possible respiratory compromise.
- **GCS < 13** → indicates reduced consciousness and possible neurological compromise.

## Key Design Principle

The model is designed as a **single-trigger early warning system**, where any one abnormal vital sign is sufficient to classify a patient as at risk. This approach prioritises sensitivity over specificity to support early identification of clinical deterioration across multiple physiological systems.

The selected variables, carried forward from the cleaned dataset in Assignment 3, were chosen because they represent independent and clinically meaningful indicators of major organ systems, including cardiovascular, respiratory, neurological, and thermoregulatory function. These measures are routinely used in clinical early warning systems and provide rapid, actionable indicators of patient status.

Mean Arterial Pressure (MAP) was excluded from the operational rule because it is a derived measure calculated from SBP and DBP. Since blood pressure is already directly represented, including MAP would introduce redundancy without adding independent clinical value.

A limitation of this approach is that its binary structure may increase false-positive rates and does not account for differing severity weights between variables. However, it is appropriate for an initial screening tool designed for early risk detection rather than precise diagnostic classification.